2. REQUIREMENT ANALYSIS

Automating Employee Data Import & Relationship Mapping Using Import Sets and Dot Walking

1. DOCUMENT PURPOSE & OBJECTIVES

This document constitutes the formal Requirement Analysis for the ServiceNow Automation Project. It builds directly upon the scope, stakeholder identification, and high-level requirements established in the preceding "1. Identification Phase" document, translating those foundational statements into precise, actionable, and traceable requirements that must be satisfied by the delivered solution.

The purpose of this document is to define, in sufficient detail, the functional behaviour the system must exhibit, the non-functional standards to which it must conform, the business rules and data logic it must enforce, and the acceptance criteria against which delivery will be measured. It encompasses all activities described in the project technical notes, including the configuration of Import Sets, the creation and execution of Transform Maps, field-level mapping between staging and target tables, and the configuration of the Incident form to display related data via dot-walking.

This document is intended to serve as the primary input into the subsequent Design Phase and should be regarded as the baseline requirements specification for all configuration, testing, and sign-off activities. All requirements are expressed in terms of system capability and expected behaviour; implementation details and specific configuration approaches will be addressed in the Design Phase documentation.

The scope of this analysis is strictly bounded by the project as defined in the Identification Phase. No requirements have been introduced that fall outside the defined scope of data import, relationship mapping, and Incident form configuration. Traceability references are provided for each functional requirement to indicate the corresponding project activity from which the requirement is derived.

2. FUNCTIONAL REQUIREMENTS

FR-01 External Data Ingestion	
Priority	Must Have
Category	Data Input
Reference	Project Activities 1 & 2
• The system must provide an administrative interface through which an authorised user can upload an externally generated employee data file for processing within the platform.	
• The interface must accept files in standard tabular formats, including Comma-Separated Values (.csv), Microsoft Excel (.xlsx), and Extensible Markup Language (.xml).	
• Upon successful file upload, the system must initiate the creation of an Import Set record, uniquely identifying the batch of data submitted for processing.	
• The system must display the imported records within a reviewable staging view, allowing the administrator to confirm that the source data has been received and loaded accurately before any transformation is performed.	

Traceability: *Project Activity 1 (initiate import via Import Sets module) and Activity 2 (load file and confirm record count).*

FR-02 Staging Table Creation	
Priority	Must Have
Category	Data Architecture
Reference	Project Activity 2.5
• The system must support the creation of a dedicated Import Set staging table to serve as an intermediary repository for raw imported data prior to transformation.	
• The staging table must be named in accordance with the data set it represents (e.g., 'import employee') and must persist independently of the source file once the import is complete.	
• The staging table must preserve all source field values in their original form, without alteration, until the Transform Map is explicitly executed.	

Traceability: *Project Activity 2.5 (staging table auto-created from uploaded file; confirmed via table inspection prior to transform configuration).*

FR-03 Data Transformation & Field Mapping

Priority	Must Have

FR-05 Incident Form Configuration via Dot-Walking

Priority	Must Have
Category	User Interface Configuration
Reference	Project Activities 1.7–1.12

- The system shall allow an administrator to modify the layout of the Incident form to include additional fields not natively present in the default form configuration.
- The system must support the addition of fields to the Incident form that are derived from related records via dot-walking — specifically, fields accessible through the reference relationship established by the 'Assigned to' field on the Incident record.
- The following dot-walked fields must be configurable for display on the Incident form: the Department of the assigned user, the Email address of the assigned user, and the Manager associated with the assigned user.
- Once added to the form layout, these fields must automatically populate with values retrieved from the related User record [sys_user] when a user is selected in the 'Assigned to' field on any Incident record.
- The dot-walked fields must be presented as read-only attributes on the Incident form; they must not be independently editable from within the Incident record.
- The form layout configuration must be persistable and apply consistently across all new and existing Incident records viewed by users with access to the configured form view.

- The system must provide a mechanism by which an administrator can initiate the execution of a configured Transform Map against a selected Import Set staging table.
- Upon execution, the system must process each staged record in accordance with the defined field mappings, inserting or updating the corresponding records in the target User table [sys_user].
- The system must communicate the outcome of the transformation to the administrator, including a status indicator confirming whether the transformation completed successfully (e.g., 'Transform state is completed').
- The system must ensure that reference field relationships — including the manager-to-employee linkage — are established correctly in the target table records as a result of the transformation.

Traceability: *Project Activity 1.7 (access form layout configuration), Activities 1.8–1.12 (add dot-walked fields: department, email, manager to Incident form).*

NFR-02 Data Integrity	
Priority	Must Have
Category	Data Quality
• The transformation process must ensure that all field values are accurately transferred from the staging table to the target User table without alteration, truncation, or loss of data.	
• Reference field relationships — in particular, the manager-to-employee linkage — must be correctly established in the target records as an outcome of the transformation. A textual manager value in the source data must resolve to the appropriate system reference in the User table.	
• The system must not create duplicate User records for employee entries that already exist in the target table. Where a coalesce (matching) field is configured, the transform must update the existing record rather than insert a new one.	
• The dot-walked values displayed on the Incident form must accurately reflect the current state of the related User record at the time the Incident form is rendered.	

3.

NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality characteristics and operational standards to which the configured solution must conform. These requirements are not directly tied to specific functional behaviours but govern the manner in which the system must perform those functions.

NFR-01 Usability	
Priority	Must Have
Category	User Experience

NFR-03 Performance & Responsiveness	
Priority	Should Have
Category	System Performance
• The system must provide timely feedback on the status of the data transformation process upon completion, allowing the administrator to confirm success or identify failure without undue delay.	
• Dot-walked field values on the Incident form must populate automatically and promptly upon user selection in the 'Assigned to' field, without requiring a manual page refresh or additional user action.	

• The import and transformation of the sample data volume (four employee records as specified in the acceptance criteria) must complete within a timeframe consistent with standard ServiceNow platform performance expectations for native Import Set operations.

NFR-04 Maintainability & Reusability	
Priority	Should Have
Category	Operational Efficiency
• The Transform Map configuration, including all field mappings, must be persistable within the platform so that it can be reused for subsequent data import cycles without requiring reconfiguration.	
• The naming conventions applied to the staging table (e.g., 'import employee') and the Transform Map (e.g., 'Test') must be descriptive and consistent, enabling future administrators to identify and reuse the configuration without reference to this document.	
• The Incident form layout configuration must be stable and not require reapplication following routine platform updates or upgrades within the same major ServiceNow release.	

4. BUSINESS RULES & DATA LOGIC

4.1 Dot-Walking Display Logic

When a User record is populated in the 'Assigned to' field of an Incident record, the Incident form must automatically traverse the reference relationship from the Incident to the User table [sys_user] and retrieve the following values for display:

- Department: The value stored in the 'department' field of the referenced User record.
- Email Address: The value stored in the 'email' field of the referenced User record.

- Manager: The value derived by traversing the 'manager' reference field on the User record to retrieve the name of the associated manager user.

These values must be displayed on the Incident form in read-only format. They must not be stored as independent data attributes on the Incident record itself; instead, they must be retrieved dynamically from the referenced User record each time the form is rendered. This ensures that any subsequent updates to the User record are automatically reflected on associated Incident forms without requiring data remediation.

4.2 Manager Relationship Mapping Rule

When an employee record is processed during the data transformation, the text value present in the source 'manager' column must be resolved against existing User records in the [sys_user] table. The resolution must identify the correct user record corresponding to the named manager and establish a system-level reference link in the imported employee's 'manager' field.

This rule has the following dependencies and implications:

- The manager record must exist within the [sys_user] table at the time the employee record's transformation is executed. If the manager is not present, the reference will fail to resolve.
- Where a batch import includes both manager and employee records, the processing sequence must result in manager records being present in the target table before dependent employee records are processed.
- The relationship established is a standard ServiceNow reference field link; it is not a flat-text copy of the manager's name and must be resolved to the system-internal record identifier.

4.3 Import Set Staging Retention Rule

The Import Set staging table must retain a complete record of the imported source data following the execution of the Transform Map. The staging table must not be automatically purged upon transformation completion, as it serves as an audit reference for the data that was imported in a given batch. Administrators may choose to review or archive staging data in accordance with operational practice, but this is outside the scope of the current project.

5. USER STORIES

US-01

As a **HR Administrator**, *I want to* upload a CSV file containing new hire details and have those records automatically added to the ServiceNow User table, *so that* new employees

are available in the system promptly and accurately without the need for manual, record-by-record data entry, reducing administrative overhead and the risk of keying errors.

US-02

As a **Service Desk Agent***, I want to* see the department, email address, and reporting manager of the caller or assigned user displayed automatically on the Incident form when I select that user*, so that* I can immediately understand the organisational context of the affected individual and make informed decisions about escalation, communication, and incident prioritisation without leaving the Incident record.

US-03

As a **ServiceNow Administrator***, I want to* configure a Transform Map once and reuse it for future import cycles without rebuilding the field mapping configuration*, so that* the ongoing operational cost of importing new employee data is minimised and the process remains consistent and reliable across multiple import events.

6. DATA DICTIONARY & FIELD MAPPING SPECIFICATIONS

Source Field (Staging Table)	Target Field (sys_user)	Mapping Type	Notes & Constraints
name	**name [name]**	Direct	Auto-mapped based on matching field names. Accepts the full display name of the employee as provided in the source file.
email	**email [email]**	Direct	Auto-mapped. Corporate email address; recommended for use as a coalesce field to prevent duplicate record creation on re-import.
department	**department [department]**	Direct	Auto-mapped. Source value must correspond to an existing Department record in the platform for the reference to resolve correctly.
phone number	**phone [phone]**	Direct	Auto-mapped where field name alignment is confirmed. Source column header may require verification against the platform field name.
manager	**manager [manager]**	Relationship	Requires manual mapping configuration. The source text value must be resolved to an existing [sys_user] record via a reference lookup defined in the Transform Map field mapping.

Note: Fields marked with a Mapping Type of "Relationship" require explicit configuration within the Transform Map to establish the correct reference linkage in the target table. Auto-mapping alone is insufficient for these fields and manual mapping steps are mandatory during the Build phase.

7. ACCEPTANCE CRITERIA

The following acceptance criteria define the conditions that must be met for the delivered solution to be considered complete and fit for purpose. Each criterion is directly traceable to one or more functional requirements and must be verifiable through a defined test or inspection activity during the User Acceptance Testing phase. The solution will not be accepted for production deployment unless all criteria marked 'Must Pass' are satisfied.

ID	Acceptance Criterion
AC-01	An authorised administrator must be able to upload the specified employee data source file and have it result in the creation of a named Import Set record and a corresponding staging table (e.g., named 'import employee') containing the records from the file, with all source columns preserved.
AC-02	A Transform Map must exist within the platform, configured to link the 'import employee' staging table as the source to the User table [sys_user] as the target. The Transform Map must contain field mappings covering all five source fields defined in the Data Dictionary (Section 6).
AC-03	Executing the Transform Map against the sample data import must result in the successful insertion of new User records in the [sys_user] table corresponding to the employees in the source file. The transformation must complete with a status of 'Completed', and the resulting user records must be visible and correctly populated in the platform.
AC-04	The manager reference field on each imported employee User record must correctly resolve to the corresponding manager's User record in [sys_user]. Inspecting the employee record must reveal a populated manager reference, not a blank or unresolved value.
AC-05	The Incident form layout must include the three dot-walked fields: Assigned to Department, Assigned to Email, and Assigned to Manager (or equivalent dot-walked field labels). These fields must be visible on the form without requiring additional navigation.
AC-06	When a new Incident record is created or edited and a user from the imported employee list is selected in the 'Assigned to' field, the Department, Email, and Manager fields on the Incident form must automatically populate with values derived from that user's record in [sys_user], without any manual input from the service desk agent.